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REVIEW ARTICLE

Factors influencing the effectiveness of surgical therapy of peri-implantitis

Ausra Ramanauskaite[1] | Wenije Zhou[2] | Emilio A. Cafferata[1,3] | Frank Schwarz[1]

[1]Department of Oral Surgery, Implantology and Oral Medicine, Goethe University, Frankfurt am Main, Germany

[2]Second Dental Center, Ninth People's Hospital, Shanghai Jiao Tong University, School of Medicine, Shanghai, China

[3]Oral Peri-Implant Research Group, Department of Oral Implantology, Postgraduate School of Dentistry, Universidad Científica del Sur, Lima, Peru

Correspondence
Ausra Ramanauskaite, Department of Oral Surgery, Implantology and Oral Medicine, Goethe University, Frankfurt am Main, Germany.
Email: ramanauskaite@med.uni-frankfurt.de

Abstract

Background: Surgical therapy is required in the majority of peri-implantitis cases due to the inconsistent efficacy of non-surgical approaches. Although surgical treatment generally offers higher predictability, reported success rates vary widely.

Aim: To review the available evidence on the factors influencing the outcomes of peri-implantitis surgical therapy, including surgical technique, implant-site characteristics, and patient-related conditions.

Materials and Methods: This narrative review is based on a literature search conducted in three electronic databases to identify relevant studies published between January 2010 and November 2025.

Results: Surgical factors: The adjunctive use of reconstructive measures improves radiographic defect fill and may help maintain soft-tissue levels, although no clear benefits have been demonstrated in the resolution of clinical signs of inflammation compared with non-reconstructive approaches. Contained intrabony defects (3- and 4-wall) consistently demonstrate more favorable outcomes of reconstructive therapy compared with non-contained defects (1- and 2-wall). Reconstruction protocols that incorporate slowly resorbing bone substitutes may lead to more favorable outcomes compared with autogenous bone. No definitive conclusions could be made regarding the optimal flap design, postoperative healing mode (i.e., submerged versus non-submerged), or the benefits of adjunctive membranes, systemic antibiotics, or additional use of biologics. Overall, the effectiveness of surgical peri-implantitis treatment is associated with the surgeon's level of experience. Implant-site-related factors: Initial disease severity characterized by extensive bone loss, suppuration, deep probing depths, rough implant surfaces, and persistent bleeding after surgery have been identified as major factors compromising treatment success. Conversely, the presence of keratinized mucosa appears to enhance the outcomes of reconstructive peri-implantitis therapy. Emerging evidence points to an association between macrophage polarization and the outcomes of surgical peri-implantitis treatment. Patient-related factors: Current evidence underscores the essential role of patient adherence to supportive postoperative care in maintaining the results achieved

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surgically. Findings concerning the influence of a history of periodontitis, smoking status, and gender remain inconsistent.

Conclusions: Outcomes of surgical peri-implantitis treatment are determined by a complex interplay of defect morphology, disease severity, implant surface characteristics, surgical expertise, and patient adherence to supportive care.

Clinical Relevance: Clinically, careful case selection, defect-specific surgical planning, and strict postoperative maintenance are essential to enhance predictability and long-term stability.

KEYWORDS
Peri-implantitis, surgery, therapeutic outcomes, treatmet success

1 | INTRODUCTION

Owing to the inconsistent efficacy of non-surgical approaches for peri-implantitis management, surgical intervention is required in the majority of cases.[1] Following implant surface decontamination, several surgical strategies can be employed, including non-reconstructive therapy, resective procedures involving hard- and/or soft-tissue removal, and reconstructive techniques utilizing bone grafting procedures.[2,3] The choice of surgical approach is largely determined by the morphology of the peri-implant bone defect.[4] Accordingly, more favorable outcomes of reconstructive therapy have been reported in the presence of a contained defect configuration (i.e., supported by 3- to 4-walls) compared with non-contained defects.[5,6]

The primary objective of surgical peri-implantitis treatment is to re-establish peri-implant tissue health and ensure the long-term stability of the implant-surrounding hard and soft tissues.[7,8] Reconstructive procedures additionally aim to regenerate lost bone, promote re-osseointegration, and maintain stable peri-implant soft-tissue levels postoperatively.[9] According to current clinical practice guidelines, treatment success—regardless of the therapeutic modality—is defined by the absence of inflammation (bleeding on probing (BOP) at ≤1 site and no suppuration (Sup) together with shallow probing depths (PD ≤5 mm)).[7] It should be acknowledged, however, that rigidly applying a PD cut-off of 5 mm can be misleading in certain clinical situations, as deeper PD values (6–7 mm) may be compatible with peri-implant health in cases with deep implant insertion or thick supracrestal soft tissues.[10]

Overall, surgical therapy for peri-implantitis is consistently associated with considerable improvements in clinical signs of inflammation (reduced BOP, PD, and suppuration), though no particular surgical modality has demonstrated clear superiority.[1,11–13] Reconstructive approaches, however, tend to yield more favorable outcomes in terms of radiographic defect fill and postoperative soft-tissue stability.[1,11,13] Reported treatment success rates, based on different success criteria, vary considerably—ranging from 33% to 75% of patients for non-reconstructive approaches and 11% to 66.6% of patients for reconstructive approaches over follow-up periods of 1–7 years.[2] Despite initial improvements, recurrence or persistence of inflammation necessitating additional surgical interventions or even implant removal is not uncommon.[2,11–12] Multiple patient- and implant-related factors have been identified as determinants of surgical peri-implantitis treatment outcomes. Therefore, the aim of this narrative review is to provide an overview of the factors influencing the effectiveness of surgical therapy for peri-implantitis.

2 | LITERATURE SEARCH STRATEGY

A comprehensive literature search was conducted for this narrative review using electronic databases (PubMed, EMBASE, and the Cochrane Library) to identify relevant studies published between January 2010 and November 2025. A combination of Medical Subject Headings (MeSH) and free-text terms was used:

Search terms

("Peri-implantitis" OR "Periimplantitis" OR "Peri implantitis" OR "Peri-implant disease")

AND ("Surgical treatment" OR "Surgery" OR "Surgical therapy" OR "Surgical debridement"

OR "Reconstructive therapy" OR "Regenerative" OR "Resective" OR "Implantoplasty")

AND ("Effectiveness" OR "Outcomes" OR "Clinical outcomes" OR "Success" OR "Implant survival")

2.1 | Inclusion and exclusion criteria

Randomized controlled trials (RCTs), prospective controlled clinical trials (CCTs), and prospective or retrospective case series with a follow-up period of at least 6 months reporting on the association between local, patient-related, and surgery-related factors and the outcomes of surgical peri-implantitis treatment were included. Studies with a minimum of 10 patients (five per group in controlled studies) were considered. All surgical treatment approaches, including non-reconstructive/resective and reconstructive techniques, as

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well as studies pooling different therapeutic modalities, were considered. The search was limited to human studies. Case reports, expert opinions, preclinical studies, and articles not available in English were excluded.

The initial electronic search identified 916 records (353 in PubMed, 281 in EMBASE, and 282 in the Cochrane Library). After screening titles and abstracts, 76 articles were selected for full-text evaluation, and 54 articles reporting on 39 original studies were included in this review. All included studies are summarized in Table S1 and were categorized according to the surgical treatment modality:

a. Studies reporting on non-reconstructive/resective therapy: $n = 15$ articles/12 studies—including 4 RCTs, 6 retrospective, and 2 prospective clinical studies.
b. Studies reporting on reconstructive therapy: $n = 34$ articles/22 studies – including 13 RCTs, 3 retrospective, and 6 prospective clinical studies.
c. Studies pooling different surgical treatment modalities: $n = 5$ articles/5 studies—including 5 retrospective studies.
d. From the included studies, 4 RCTs directly compared reconstructive versus non-reconstructive treatment approaches.

3 | SURGICAL FACTORS

3.1 | Flap design

Flap design during surgical peri-implantitis treatment should provide adequate access to the defect while minimizing surgical trauma and postoperative discomfort for patients. Table S1 summarizes the incision techniques and flap designs used in the included studies.

For the non-reconstructive/resective treatment modalities, the majority of studies employed a conventional full-thickness access flap, typically a mid-crestal incision often combined with relieving vertical incisions to ensure adequate visibility and instrumentation of the defect. In most cases of non-reconstructive surgery, the flaps were repositioned apically or sutured to allow pocket elimination and secondary intention healing of soft tissues.

For reconstructive therapies, most studies likewise utilized full-thickness flaps with broad access, aiming for primary wound closure over the augmented defect. Flaps were generally repositioned to cover the regenerative materials and sutured for tension-free primary closure (often requiring periosteal releasing incisions). Notably, 3 studies did not clearly specify the exact flap design utilized.

In periodontitis therapy, there has been a focus on minimally invasive interventions that limit flap elevation to reduce surgical trauma and enhance flap stability. Such strategies include papilla-preservation techniques, the entire papilla-preservation flap, and other designs intended to restrict flap extent.[14–16] Among the included peri-implantitis studies, no studies explicitly reported the use of a minimally invasive flap design. It should be pointed out that compared with periodontitis, peri-implantitis lesions are often significantly larger and frequently require circumferential access to adequately debride and decontaminate the implant surface.[17] Thus, in practice, a greater flap elevation is typically needed around peri-implantitis sites than around periodontitis sites.

Interestingly, a novel flapless surgical approach has been explored recently as an alternative to traditional flap elevation. In a large prospective case series including 117 patients (338 implants), a flapless surgical access (pocket curettage without raising a flap) led to disease resolution at 54.4% of implants over a follow-up period of 5 years.[18] While this flapless approach achieved only moderate success, it suggests that less invasive surgical strategies might be viable in certain scenarios, provided that rigorous decontamination and maintenance are ensured. Nonetheless, controlled studies are needed to validate the efficacy of such minimally invasive or flapless techniques.

3.2 | Efficacy of reconstructive versus non-reconstructive therapy

Several studies have compared non-reconstructive (open-flap debridement or resective) approaches to reconstructive (regenerative) approaches in the surgical management of peri-implantitis. According to four RCTs that directly compared these modalities, treatment success (based on varying composite success definitions) over 12 months to 3 years ranged from 13.5% to 85% of implants treated with reconstructive therapy, and from 5% to 90% of implants treated with open-flap debridement (Table S1; Figure 1).[19–24] For example, in two 12-month RCTs, reconstructive groups showed significantly greater radiographic defect fill, whereas clinical parameters (reductions in BOP, PD, and mid-buccal recession) were comparable between reconstructive and open-flap groups.[20,22] One RCT did report a higher postoperative soft-tissue level (less recession) at 12 months when reconstructive measures were used, but this benefit was not maintained at 3 years.[23,24]

These findings are corroborated by meta-analyses indicating that adjunctive reconstructive measures yield significantly greater radiographic defect fill than open-flap debridement alone, while achieving similar reductions in inflammation (BOP and PD).[1,11,13] In terms of soft-tissue level changes, one meta-analysis found that reconstructive therapy conferred an advantage in maintaining postoperative soft-tissue levels compared with non-reconstructive approaches (weighted mean difference in recession: −1.35 mm in favor of regeneration).[25]

In summary, reconstructive surgical therapy tends to improve hard-tissue outcomes (bone fill) and may better preserve soft-tissue levels, but it does not appear to result in superior resolution of peri-implant inflammation compared with open-flap debridement alone. Thus, when choosing between reconstructive or resective approaches, clinicians must weigh the goal of defect fill against the complexity and cost of reconstructive procedures, acknowledging that improvements in clinical parameters (pocket depth reduction, bleeding elimination) are likely to be similar between the approaches.

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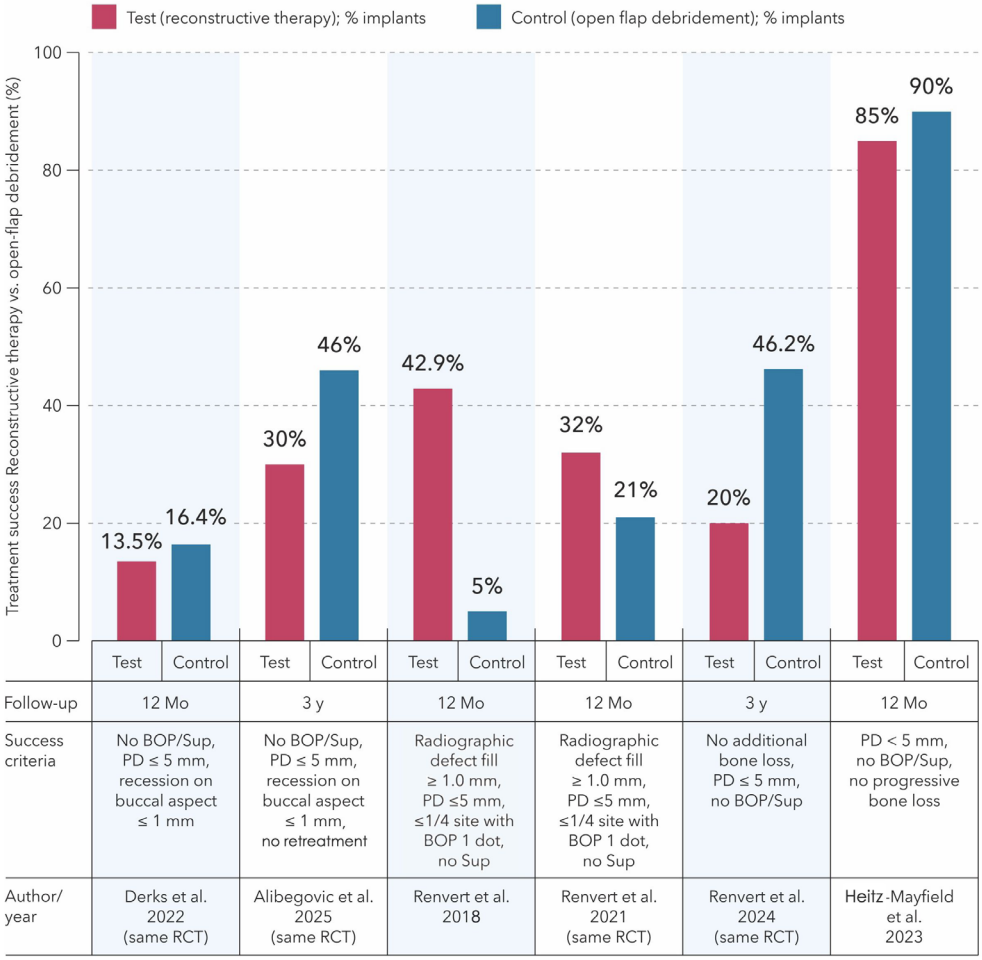


FIGURE 1 Treatment success of reconstructive therapy vs. open-flap debridement reported by comparative studies using combined success criteria.

3.3 | Biomaterials and biologics

3.3.1 | Reconstructive therapy

Table S1 provides an overview of the reconstructive protocols (bone grafts and membranes, with or without biologic adjuncts) used to treat peri-implantitis-related bone defects in the included studies.

Choice of bone graft: The type of bone graft material can influence outcomes. One comparative clinical study with 3–5 years follow-up reported superior outcomes (greater PD and BOP reduction and more radiographic fill) for guided bone regeneration (GBR) using a xenogenic particulate bone filler (deproteinized bovine bone mineral, DBBM) compared with autogenous bone chips.[5,26] Over 4 years, a GBR approach with xenogeneic bone filler also led to better PD and BOP reductions than an alloplastic bone filler alone.[27] Additionally, two different xenogeneic fillers (bovine hydroxyapatite vs. DBBM) showed similar clinical and radiographic outcomes at 12 months, suggesting both xenografts were comparably effective.[28] In another study, using titanium granules versus xenogenic bone particles, the titanium granules yielded significantly greater radiographic defect fill, although clinical outcomes (PD and BOP changes) were similar.[29]

Emerging evidence suggests that slowly resorbing bone substitutes (such as xenografts) may be preferable to faster-resorbing autografts in peri-implantitis defects. A recent network meta-analysis (based on 12-month data from five studies) indicated that protocols using xenogeneic bone achieved significantly greater resolution of inflammation (larger reductions in BOP and PD, and less recession) than those using autogenous bone.[30] In light of these findings, reconstruction protocols incorporating xenogeneic or other low-resorption grafts may lead to more favorable medium-term outcomes than those using autogenous bone. However, long-term comparisons are still needed, and defect morphology and patient factors may also play critical roles.

Use of biologics: There is limited evidence regarding the added benefit of biologic agents (e.g., enamel matrix derivative (EMD), growth factors, platelet concentrates) in reconstructive peri-implantitis therapy. One recent RCT evaluated EMD as an adjunct to regenerative surgery with xenogenic bone and a collagen membrane.[31] At 12 months, sites treated with EMD showed similar PD reductions and success rates as sites treated without EMD, suggesting no significant clinical advantage to adding EMD.[31] Similarly,

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over 5 years, the adjunctive use of EMD resulted in similar outcomes when compared with open-flap debridement alone.[32,33]

The efficacy of platelet-rich fibrin (PRF) was tested in one RCT: at 6 months, PRF-treated sites had modestly greater PD reduction, CAL gain, and increase in keratinized tissue compared with open-flap debridement, indicating a potential benefit.[34] Another RCT investigated adjunctive hyaluronic acid with xenogenic bone.[35] While radiographic defect fill improved with hyaluronic acid, there were no differences in PD or BOP outcomes compared with the control group.[35] Overall, due to the limited and short-term data, no definitive conclusions can be drawn on the benefits of adding biologic agents to reconstructive treatments.

3.4 | Significance of a barrier membrane

The principle of GBR involves placing a barrier membrane over the graft material thereby creating a space for bone regeneration.[36] A membrane can also help stabilize the graft and maintain the augmented contour. Generally, using a membrane should improve regenerative outcomes in peri-implantitis defects, particularly in non-contained defects where graft stability is a challenge.

Summary of the findings: The clinical significance of adding a barrier membrane in peri-implantitis reconstructive therapy remains controversial. One 4-year study reported significantly greater BOP and PD reductions when a xenogenic bone graft was combined with a native collagen membrane, compared with sites treated with a graft alone. In contrast, multiple investigations found no additional clinical benefit from membrane use, whether native or cross-linked collagen membranes were placed over allografts or xenografts, up to 5 years postoperatively.[37–40] For example, a 12-month RCT using a demineralized allograft + membrane vs. the same graft without a membrane showed similar success rates (i.e., ≤1 spot of BOP and/or Sup, PD ≤5 mm, no radiographic progressive bone loss; about 75% vs. 79%) in the two groups.[37] Another 3-year study using a collagen-stabilized xenograft reported 17.6% vs. 27.8% success with vs. without a membrane at 1 year, and 36.8% vs. 45% at 3 years with no statistically significant difference between groups.[40] Interestingly, in that RCT the membrane group experienced more postoperative pain and complications (e.g., 19% soft-tissue dehiscence, 9.5% membrane exposure, 4.8% graft exposure) than the no-membrane group.[39]

It should be noted that many of these studies pooled various defect configurations (2-, 3-, and 4-wall defects) in their analysis. It is plausible that in partially contained defects (1- or 2-wall scenarios), the membrane plays a more critical role in preventing soft-tissue collapse and maintaining the graft, whereas in inherently contained 4-wall defects, a membrane might confer little added benefit. In summary, current data do not show a clear overall clinical benefit to using barrier membranes in reconstructive peri-implantitis therapy, especially given the potential for increased patient morbidity. Membranes may be more justified in non-contained defects, but further stratified research is needed.

3.5 | Defect configuration

3.5.1 | Reconstructive therapy

Peri-implantitis lesions can present various bone defect morphologies. Commonly, defects are classified as Class I (intrabony defects with a surrounding bony wall), Class II (supracrestal horizontal bone loss), or Class III (combined intrabony and supracrestal defects).[41] Clinical data indicate that the majority of peri-implantitis sites (around 79%) are combination defects (Class III).[41] Pure intrabony lesions often show circumferential bone loss with intact buccal and lingual walls (true 4-wall circumferential defects, observed in about 55% of intrabony cases), followed by 3-wall defects (e.g., facial dehiscence-type defects).[41] Pure horizontal defects (Class II) are relatively uncommon (as low as about 2% of lesions) (Figure 2).[41]

Figure 3 represents details on the defect configurations reported in the included clinical studies of peri-implantitis surgery. Overall, 16 studies did not provide any information on the defect configuration of treated sites. Several studies have identified defect morphology as a key determinant of reconstructive treatment outcomes. In particular, contained defects supported by more bony walls consistently yield better regenerative results:

- In an RCT, reconstructive therapy with an anorganic bovine bone and collagen membrane achieved significantly greater PD reduction and clinical attachment gain at 4-wall defects compared with 2- or 3-wall defects after 6 months.[6]
- Similarly, at 12 months, another study found that 4-wall defects (and deeper baseline defects) showed greater radiographic bone fill than 2- or 3-wall defects when treated with either autogenous bone or xenograft plus membrane.[42]
- A 3-year RCT reported that using bovine bone plus either a collagen membrane or concentrated growth factor membrane led to 6- to 7-fold higher odds of treatment success (PD ≤5 mm and no additional bone loss) in 4-wall defects compared with 3-wall or 2-wall defects.[43] The 7-year follow-up of this cohort confirmed significantly higher disease resolution rates (PD ≤5 mm, BOP ≤1 site, no suppuration, no retreatment or loss) in 4-wall versus 2-wall defects.[44]
- Moreover, a recent analysis demonstrated that defect angle correlates with regenerative outcome, since narrower defects (mean angle about 40°) yielded about 1.85 mm of radiographic bone gain, whereas wider defect angles were associated with less bone gain.[45]

On the other hand, not all data are uniform. A 3-year RCT using a novel volume-stable collagen matrix-embedded bone filler found no association between defect configuration and treatment outcome, and a 5-year evaluation of the same material likewise showed no significant influence of defect shape.[23,24,46] These discrepancies could be related to, for example, differences in graft materials, as volume-stable scaffolds might maintain space even

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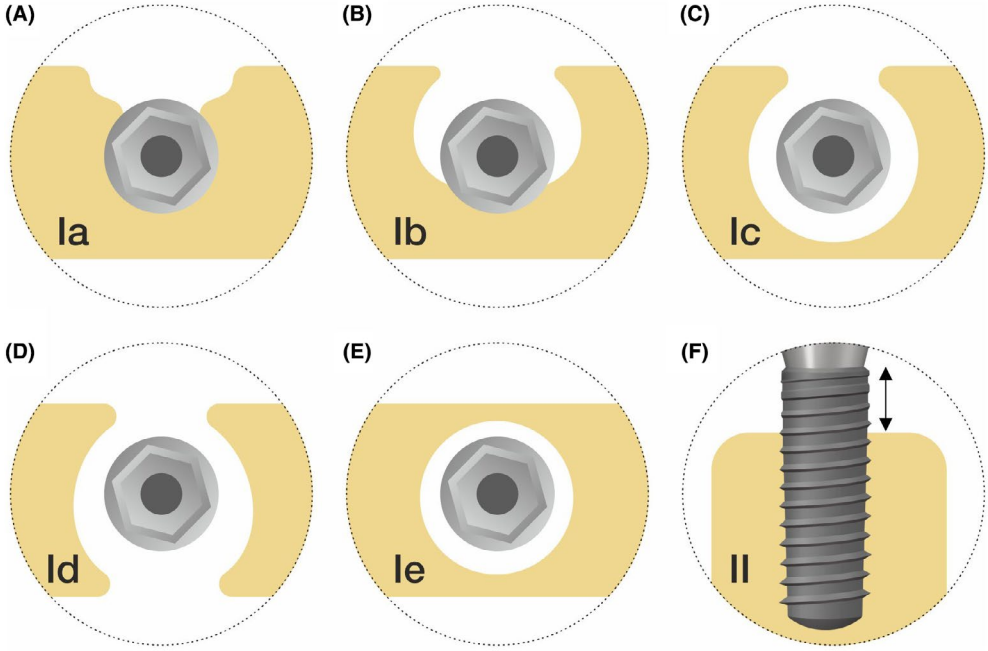


FIGURE 2 Classification of peri-implantitis-related bone defect configurations according to Schwarz et al. ([41] Occlusal view: (A) Class Ia; (B) Class Ib; (C) Class Ic; (D) Class Id; (E) Class Ie; (F) Class II–vestibular view.

in non-contained defects, whereas particulate grafts rely more on defect walls for stability. Nonetheless, from a biologic perspective, defects with three to four bony walls and at least 3-mm defect depth provide a more favorable environment for clot stabilization and revascularization than 1- or 2-wall defects. Therefore, they are considered more amenable to regeneration. Figures 4 and 5 represent the clinical cases utilizing reconstructive therapy in 4- and 3-wall defects, respectively.

Considering the defect configurations in the included reconstructive studies, the case selection criteria were often unclear or did not strictly require contained defects. Specifically, the indication criteria were either unclear in 43% of reconstructive studies, while a separate 43% of studies were inconsistent with a contained defect approach (i.e., many studies included or did not exclude non-contained defects). This highlights that in effectiveness studies, reconstructive techniques are frequently attempted in suboptimal defect morphologies, which could partially explain the variable outcomes observed (Figure 3).

3.6 | Flap adaptation and postoperative healing mode

A critical intra- and postoperative consideration is whether the surgical site is left to heal in a submerged (completely covered) or a non-submerged (transmucosal) mode. Basically, flap adaptation techniques, suturing methods, and the decision to submerge the implant during healing can influence outcomes by affecting wound stability and infection control.

In periodontal surgery, apically repositioned flaps are often used in resective treatment to reduce pockets, whereas primary closure is sought in regenerative procedures. Similar principles may also be applied at peri-implantitis sites:

- For non-reconstructive surgical therapy, clinicians may consider repositioning the flaps apically or at least ensuring they are not completely covering the implant to maximize post-surgical pocket depth reduction. This approach may facilitate an open drainage if needed and reduces the risk of deep residual pockets.
- For reconstructive therapy, achieving primary closure and, if possible, submerged healing is thought to promote more predictable bone regeneration. Submerged healing isolates the defect site from the oral environment during early healing, potentially leading to superior outcomes.

Evidence from a preclinical canine study supports submerged healing. In particular, surgically treated peri-implantitis defects that were kept submerged for 3 months had significantly higher bone-to-implant contact than those that healed with transmucosal healing.[47] Clinically, a reanalysis of two prospective studies compared outcomes of reconstructive therapy under submerged vs. non-submerged conditions.[48] In the submerged group, patients underwent 8 months of healing with the prosthesis removed and a cover screw in place (using a composite graft of allograft + xenograft + autograft plus a non-resorbable PTFE membrane). In the non-submerged group, the same graft with a resorbable membrane was used and the implants healed transmucosally with the prosthesis or a healing abutment in place.[48] The submerged-healing group demonstrated more

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(A)

	Author/years	Intrabony defect	Horizontal defect
Consistent	Ramanauskaite et al. 2025[56]	Three-to-four wall intrabony defects ≥ 3 mm, assessed radiographically and during surgery	Defects treated by means of resective therapy were excluded
	Monje et al. 2023[38]	Three-to-four wall intrabony defects ≥ 3 mm, assessed radiographically	Specified presence or absence of supracrestal bone loss
	Schwarz et al. 2010[6]	Three-to-four wall intrabony defects > 3 mm, assessed radiographically and during surgery	Supracrestal component >1 mm
Inconsistent	Regidor et al. 2025[33]	Two (mesial and distal), three-, or four-wall intrabony defects ≥ 3 mm depth and width ≤ 4 mm, assessed during surgery	Unclear
	Renvert et al. 2024[21]	Three- to four-wall intrabony crater-like defects ≥ 3 mm, assessed radiographically and during surgery	Unclear
	Regidor et al. 2023[40], 2025[41]	Two (mesial and distal), three-, or four-wall intrabony defects ≥ 3 mm, assessed radiographically and during surgery	Specified presence or absence of supracrestal bone loss
	Rakašević et al. 2023[36]	Three- to four-wall defects (Class Ib, Ic, and Ie), assessed during surgery	Supracrestal components (Class IIb) were treated by means of resective therapy
	Monje et al. 2020[119]	Two- and three-wall intrabony crater-like defects ≥ 3 mm, assessed radiographically and during surgery	Class III defects supracrestal components were treated by means of resective therapy
	Isler et al. 2018[73], 2022[75], 2024[44], 2025[45]	Two-, three-, or four-wall intrabony defects ≥ 3 mm, assessed during surgery	Specified presence or absence of supracrestal bone loss
	Renvert et al. 2018[24]	Three- to four-wall intrabony crater-like defects ≥ 3 mm, assessed radiographically and during surgery	Unclear
	Hamzacebi et al. 2015[35]	Two-, three-, or four-wall intrabony defects ≥ 2 mm, assessed radiographically and during surgery	Supracrestal component from 1 to 5 mm
	Roos-Jansaker et al. 2014[39]	One-to-four wall intrabony defects, assessed during surgery	Unclear
Unclear	Wen et al. 2024[49]	Crater-like intrabony defects ≥ 3 mm, assessed during surgery	Class II defects were excluded
	Heitz-Mayfield et al. 2023[23]	Three-, or four-wall intrabony defects ≥ 3 mm, assessed during surgery	Unclear
	Derks et al. 2022[26], 2023[67], 2025[120]	No minimum number of bony walls circumferential defects ≥ 3 mm, assessed during surgery	Unclear
	Polymeri et al. 2020[30]	Three- to four-wall intrabony defects ≥ 3 mm, assessed during radiographically	Unclear
	Roccuzzo et al. 2017[47], 2021[79]	Class I crater-like intrabony defects, assessed during surgery	Class II and III defects were excluded
	Guler et al. 2017[31]	Three-, or four-wall intrabony defects, assessed radiographically	Unclear
	Isehed et al. 2016[34], 2018x[121]	Intrabony defects ≥ 3 mm, assessed during surgery	Unclear
	Aghazadeh et al. 2012[28], 2020[43], 2022[5]	Two-to-three wall intrabony defects ≥ 3 mm, assessed radiographically and during surgery	Unclear
	Schwarz et al. 2009[29]	Intrabony defects > 3 mm, assessed radiographically and during surgery	Supracrestal component assessed

(B)

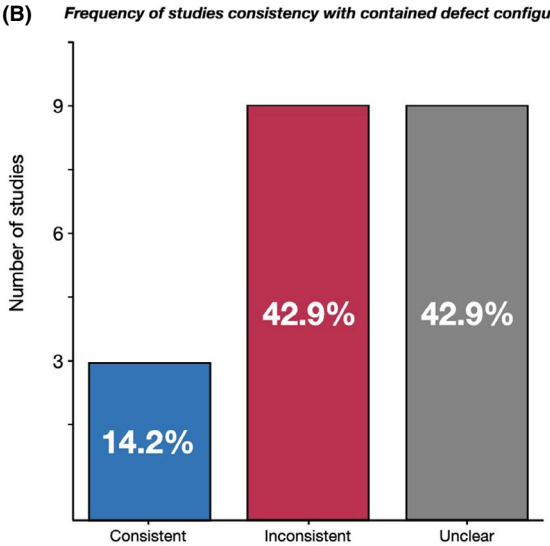


FIGURE 3 (A) Specifications of defect configurations in the studies reporting on the clinical outcomes of reconstructive peri-implantitis therapy. (B) Bar-diagram depicting the frequency of studies consistent with the contained defect configuration.

favorable results since at re-entry (about 8 months), clinical defect fill was about 0.9 mm greater, and at 12 months radiographic fill was about 1.7 mm greater than in the non-submerged group.[48] PD reduction was also higher by about 1.3 mm with submerged healing, while BOP reduction was similar between groups. These differences were statistically significant.[48]

Recognizing these trends, the recent AO/AAP consensus conference emphasized that for non-reconstructive peri-implantitis

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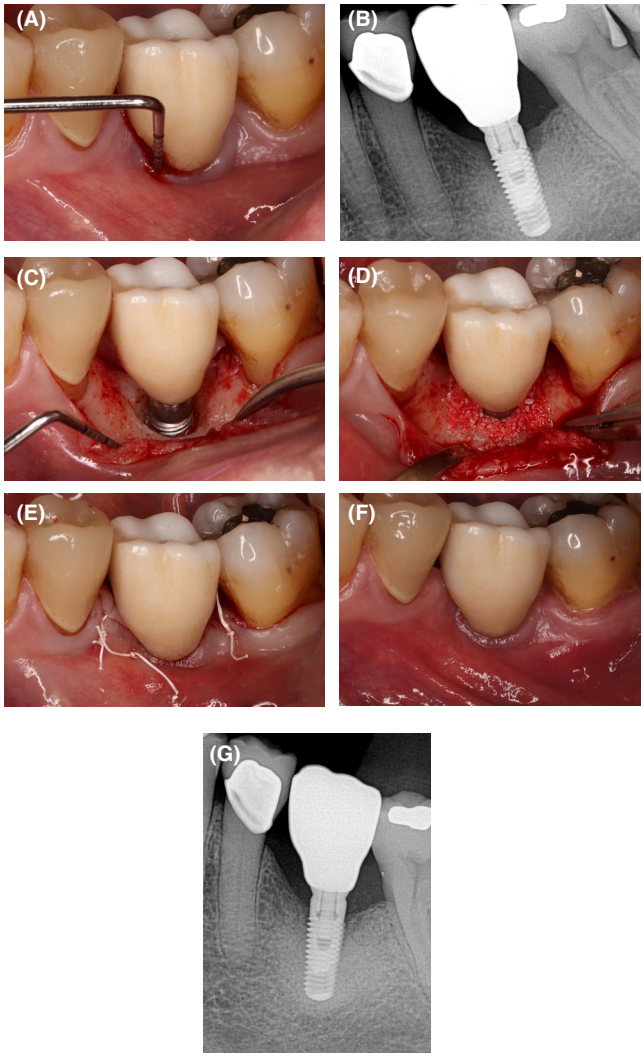


FIGURE 4 Clinical case illustrating a surgical reconstructive peri-implantitis. (A) Intraoral preoperative view showing clinical signs of inflammation at implant in region 36. (B) Radiographic view displaying marginal bone loss at implant 36. (C) Intrabony circumferential defect of 4 mm depth was detected following the access to the defect. (D) Intrabony defect homogeneously filled with a collagenated bovine bone mineral (90% bovine bone mineral + 10% porcine-derived collagen). (E) Postoperative sutures. (F) Intraoral view after 12 months showing peri-implant tissue health. (G) Radiograph after 12 months depicting radiographic improvements in marginal bone levels.

surgery, an apically positioned flap or similar repositioning may be beneficial to maximize PD reduction, whereas for reconstructive therapy, a submerged-healing approach with primary closure might enhance outcomes.[49] However, the consensus also emphasized that controlled clinical studies are needed to support these recommendations.

In this context, it is also important to weigh the practical implications of submerged healing. Achieving complete coverage often requires removing the prosthesis and possibly placing a provisional restoration, which increases treatment time, cost, and risk of prosthetic complications. Additionally, primary closure over an implant is technique-sensitive and can be prone to early wound dehiscence. Indeed, early postoperative wound dehiscence has been frequently reported with submerged healing in reconstructive peri-implantitis therapy, with one study noting dehiscences in 25 of 36 sites despite primary closure attempts.[50] Since such complications can compromise treatment outcomes, clinicians should carefully consider flap management in reconstructive therapy.

In summary, while submerged healing seems advantageous for regenerative outcomes, its benefits must be balanced against the higher complexity and complication risk. No definitive consensus has been reached on an "optimal" healing protocol, and clinical judgment is required on a case-by-case basis.

3.7 | Systemic antibiotics

Considering the bacterial etiology of peri-implantitis, systemic antibiotics have been investigated as adjunctive therapy in combination with surgical management. The rationale is that reducing the bacterial load systemically could enhance the effect of mechanical debridement and implant surface decontamination during surgery. Here we summarize the findings from studies of systemic antibiotic use in peri-implantitis surgery, including non-reconstructive and reconstructive approaches.

3.7.1 | Non-reconstructive/Resective therapy

According to one 12-month RCT, adjunctive systemic azithromycin (250 mg twice daily for 5 days, starting 1 day pre-surgery) in conjunction with open-flap surgery failed to significantly improve clinical outcomes compared with surgery alone.[51] Successful treatment, defined as PD ≤5 mm, absence of BOP and Sup, and no bone loss ≥0.5 mm, was achieved in 7/15 (46.7%) patients treated with antibiotics and 4/16 (25.0%) patients treated without antibiotics, with no statistically significant difference between groups ($p = 0.20$).[51] Furthermore, the overall bacterial load was considerably reduced after 2 and 4 months, but returned to baseline levels at later time points, with comparable results observed in both groups.[51] Interestingly, no differences in bacterial load were observed between successfully and unsuccessfully treated implants.[51]

Another RCT demonstrated significantly higher treatment success (PD ≤5 mm, no BOP/Sup, and no bone loss >0.5 mm) after 12 months at modified-surfaced implants treated with a non-reconstructive/resective approach with adjunctive systemic antibiotics (amoxicillin 750 mg twice daily for 10 days; OR:38.69; $p = 0.005$).[52,53] The observed benefit of systemic antibiotics was, however, not sustained over the 3-year period.[52,53] For implants with non-modified surfaces, systemic antibiotics did not provide significant advantages at any time point.[52,53] More recently, a three-armed RCT assessed the effects of systemic antibiotics as an adjunctive measure to resective peri-implantitis treatment.[54] After

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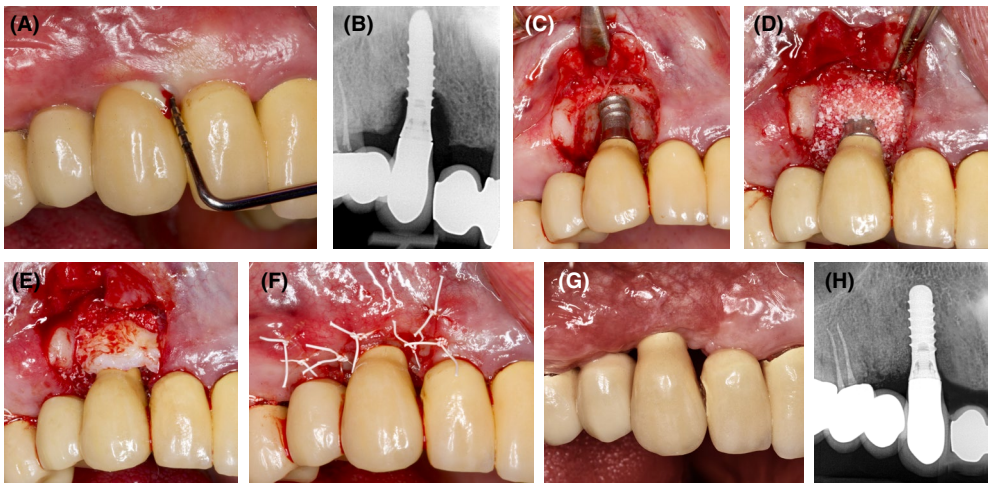


FIGURE 5 Clinical case depicting reconstructive surgical procedure employing 90% bovine bone mineral +10% porcine-derived collagen in region 13. (A) Presence of profuse bleeding on probing and suppuration along with probing depth values up to 9 mm at implant 13. (B) Radiographical presence of marginal bone loss at implant 13. (C) Access to peri-implant bone defect; evidence of a 3-wall intrabony defect configuration. (D) Reconstruction of a defect with collagenated bovine bone mineral. (E) Simultaneous soft-tissue volume grafting using a connective tissue graft to compensate soft-tissue remodeling following the healing. (F) Suturing of the tissues. (G) Intraoral view after 12 months showing healthy tissue conditions. (H) Radiograph depicting fill of the defect 12 months after the treatment.

12 months, administration of a 7-day course of either amoxicillin + metronidazole (Test 1) or phenoxymethylpenicillin + metronidazole (Test 2) failed to improve clinical outcomes compared with placebo, as indicated by similar changes in PD, BOP, and Sup across all groups.[54] However, radiographic marginal bone stability and a subsequent disease resolution (PD ≤5 mm, BOP ≤1 site, no Sup, and no additional bone loss >0.5 mm) were significantly more frequent in the antibiotic groups (Test 1: 68% of implants; Test 2: 66%) compared with placebo (28%).[54]

3.7.2 | Reconstructive therapy

One single retrospective analysis has focused specifically on systemic antibiotics administered with reconstructive therpay.[55] Among 49 patients, 27 received a single preoperative dose of amoxicillin (2 g), 12 patients received postoperative antibiotics for 3 days, and 10 patients did not receive any systemic antibiotics.[55] Overall, after 12 months, reconstructive treatment resulted in considerable reduction in BOP and PD values that were comparable among the groups. Disease resolution (PD ≤5 mm, ≤ 1 BOP site, no Sup) after 12 months was observed in 61.5%, 73.7%, and 89.5% of patients in the no-antibiotic, preoperative, and postoperative groups, respectively, with no detected significant differences among the groups.[55]

3.7.3 | Pooled various surgical therapies

When considering various surgical treatment approaches, one retrospective analysis found the prescription of systemic antibiotics either preoperatively (2 g of amoxicillin) or pre- and postoperatively to be significantly associated with higher rates of treatment success after 12 months (i.e., max PD ≤5 mm, ≤ 1 BOP site, and no Sup) (multivariate analysis: preoperative: OR:3.54, $p = 0.04$: pre- and postoperative: OR:4.49, $p = 0.02$).[56] Similarly, the estimations of a recent meta-analysis reported significantly higher short-term (1–2 years) treatment success (based on varying definitions; 5 studies; OR:2.73), increased bone level stability (3 studies; OR:2.73), reduced BOP (3 studies; OR:0.33), and reduced Sup (3 studies; OR:0.33) following systemic antibiotic administration along with the surgical peri-implantitis treatment.[57] However, when interpreting these outcomes, it is important to note that the analysis pooled studies employing heterogeneous surgical approaches (non-reconstructive, resective, and reconstructive therapies) and different antibiotic regimens, which all might confounded the results. Nonetheless, considering the risks of superinfections, microbiome dysbiosis, and antibiotic resistance, current clinical practice guidelines do not recommend the routine use of systemic antibiotics in the treatment of peri-implant diseases.[7]

3.8 | Operator-related factors

Among the included studies, 20 specified the experience level of the surgeons.

3.8.1 | Pooled various surgical therapies

A recent retrospective analysis conducted at a single university center evaluated the efficacy of surgical peri-implantitis treatment performed by four experienced oral surgeons. Despite a high degree of

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procedural standardization, there was significant variability in treatment success rates between the four operators after 12 months. The procedures performed by less experienced surgeons were associated with lower success rates (multivariate analysis: OR = 0.12, $p = 0.003$).[56] These findings suggest that experience, individual surgical skills, and intraoperative decision-making may also affect clinical outcomes.

3.8.2 | Non-reconstructive/Resective therapy

The aforementioned observation supports the results of a prior retrospective study, which identified the operators experience level as a critical determinant of treatment success in resective peri-implantitis therapy.[58] Specifically, patients treated earlier within the observation period experienced significantly higher failure rates compared with those treated later, indicating the development of a learning curve associated with the surgical management of peri-implantitis.[58] Furthermore, another retrospective analysis reported that when two implants were surgically treated for peri-implantitis within the same session, employing either non-reconstructive or reconstructive approaches, the odds of treatment failure (i.e., radiographic bone loss >1.0 mm, the need for surgical retreatment, or implant loss/removal) were significantly increased (multivariate analysis: OR = 31.68, $p = 0.001$) over a mean follow-up of 3 years.[59]

Taken together, surgeon-related factors, including experience, intraoperative decision-making process, and case complexity (e.g., number of implants treated), appear to affect the success of peri-implantitis surgery, emphasizing the importance of training, procedural consistency, and individualized case planning.

4 | LOCAL FACTORS

4.1 | Initial severity of the disease

4.1.1 | Non-reconstructive/Resective therapy

Numerous studies have highlighted the prognostic relevance of initial bone loss severity on the effectiveness of surgical peri-implantitis treatment. Specifically, 6 months after non-reconstructive therapy, 74% of implants (34/46) with initial bone loss of 2–4 mm, measured intraoperatively, were classified as healthy, compared with 50% of implants with initial bone loss of 5–6 mm.[60] Among these, implants featuring ≥7 mm of bone loss, 7 from 18 were removed by the 3-month follow-up.[60] A similar trend was reported in another 6-month prospective case series, where implants with initial bone loss exceeding 7 mm were significantly more likely to present BOP, Sup, and the combination of BOP and PD ≥4 mm or PD ≥6 mm, after non-reconstructive therapy at follow-up.[61] Indeed, none of the implants with initial bone loss ≥7 mm were free of BOP, and 39% had Sup at 6 months.[61] These findings align with a retrospective analysis showing higher success rates at 12 months after non-reconstructive peri-implantitis treatment, defined as no progressive bone loss >1 mm, absence of Sup, no BOP at ≥2 sites, and no PD > 5 mm, at implants with initial bone loss limited to 1–3 mm (70%), compared with only 25% success at those with bone loss of 7–9 mm.[58] In the same study, initial bone loss was found to be a significant predictor of treatment failure, defined as implant loss/removal or progressive bone loss ≥1 mm, Sup, BOP at 3–4 sites, and PD ≥5 mm, after 12 months (multivariate analysis: OR = 1.48; $p = 0.03$).[58]

Longer-term studies further confirmed the negative effect of initial bone loss extension upon therapeutic outcomes of surgical therapy. In a retrospective study with a mean follow-up period of 6.4 ± 2.7, baseline marginal bone loss ≥4 mm significantly increased the risk for further disease progression, defined as bone loss ≥1.0 mm or implant loss following non-reconstructive peri-implantitis therapy (multivariate analysis: OR = 5.15; $p = 0.027$).[62] Similarly, over a 7-year period, non-reconstructive peri-implantitis resulted in significantly higher implant loss rates at implants featuring initial bone loss ≥60% of implant length (hazard ratio [HR] = 4.2; 95% CI: 1.9–9.4; $p < 0.001$).[63]

4.1.2 | Pooled various surgical therapies

When considering studies that pooled different surgical treatment modalities, implants with >50% bone loss had a 23.6-fold increased risk of failure, defined as positive BOP and/or longitudinal bone loss ≥0.5 mm and/or PD ≥5 mm, compared with those exhibiting <25% bone loss (OR = 23.6; $p = 0.014$).[64] Similar outcomes were observed in a retrospective evaluation of 328 implants in 150 patients, with a mean follow-up of 26 ± 20 months after non-reconstructive peri-implantitis treatment.[65] This study reported significantly reduced treatment success rates, defined as absence of BOP, absence of Sup, and PD ≤4 mm, at implants with initial bone loss exceeding 1/3 to 2/3 of the implant length (OR = 6.5; 95% CI: 1.4–30; $p < 0.05$).[65]

4.1.3 | Non-reconstructive/Resective therapy and pooled various surgical therapies

In the same context, baseline PD and Sup have also emerged as critical determinants of success. A sub-analysis of a 12-month multicenter RCT comparing reconstructive versus non-reconstructive peri-implantitis therapy identified baseline PD as the strongest predictor of surgical treatment outcomes, regardless of the treatment modality employed.[23] Higher baseline PD was positively correlated with final PD, postoperative recession, and marginal bone gain, but negatively associated with the likelihood of pocket closure, defined as final PD ≤5 mm (OR = 0.71; 95% CI: 0.54–0.93; $p < 0.05$).[66] In addition, initial PD > 8 mm was negatively associated with resective peri-implantitis treatment success, defined as absence of BOP and PD ≥6 mm (multivariate analysis: OR = 2.1; 95% CI: 1.1–3.9; $p = 0.05$).[61] Moreover, implants displaying preoperative Sup were more likely to present persistent Sup 6 months after resective therapy (OR = 23.3; 95% CI: 1.6–333.3).[61] The presence

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of Sup prior to non-reconstructive peri-implantitis treatment was also found to be a significant predictor of implant loss over a 7-year follow-up (HR = 2.87; 95% CI: 1.89–9.39; $p = 0.007$).[63] In corroboration are the results of a recent retrospective study pooling different surgical approaches, which found the presence of Sup at baseline to be a predictor for implant loss after 12 months (multivariate analysis: OR 4.51, $p = 0.002$).[56] In fact, each 1 mm increase in maximum PD at baseline increased the need for surgical retreatment over a 12-month period by 30%.[56]

In light of the aforementioned findings, greater baseline disease severity, characterized by extensive bone loss, deeper pockets, and the presence of Sup, consistently remain as significant negative predictors of surgical outcomes of peri-implantitis treatment.

4.2 | Keratinized mucosa

Growing clinical evidence highlights the essential role of keratinized peri-implant mucosa in maintaining peri-implant tissue health.[67] Implant sites with a lack of or reduced keratinized mucosa width (<2 mm) have been consistently associated with a higher prevalence of peri-implantitis, increased plaque accumulation, soft-tissue inflammation, mucosal recession, and marginal bone loss.[68] According to a recent cross-sectional analysis, 61.3% of patients with dental implants exhibiting <2 mm of keratinized mucosa experienced discomfort during brushing, which in turn correlated with higher prevalence of peri-implant disease.[69] A 20-year prospective cohort study further highlighted the importance of keratinized mucosa at dental implants, demonstrating a greater incidence of peri-implant diseases at implant sites lacking keratinized mucosa, even among patients adhering to regular supportive therapy.[70] In addition to its role in disease prevention, keratinized mucosa also appears to influence inflammation resolution in peri-implant tissue. As shown by a prospective cohort study evaluating peri-implant mucositis therapy, odds of disease resolution at 6 months were approximately 80% lower at sites with <2 mm of keratinized tissue.[71]

4.2.1 | Reconstructive therapy

The role of keratinized mucosa on peri-implantitis treatment outcomes remains controversial. Two clinical investigations reported on the influence of preoperative keratinized mucosa width on the outcomes of surgical reconstructive peri-implantitis treatment. In particular, as shown by a 12-month RCT, the presence of preoperative keratinized mucosa width significantly increased the odds of disease resolution (i.e., no BOP/Sup + PD ≤5 mm + no radiographic bone loss ≥1 mm) after reconstructive peri-implantitis therapy applying demineralized and mineralized cortical allograft with or without membrane (OR:2.10, $p = 0.035$).[37] In fact, each additional millimeter of baseline keratinized mucosa increased the likelihood of disease resolution by 110%.[37] In line with these results, one retrospective study showed that keratinized mucosa width ≥2 mm at the end of the submerged-healing phase following reconstructive peri-implantitis therapy was significantly associated with greater radiographic defect fill throughout the 10 years follow-up period.[72] Furthermore, presence of keratinized mucosa ≥2 mm at 6 months was directly associated with treatment success after 10 years (i.e., ≤1 site with BOP, no Sup, PD reduction, no further bone loss, change in mucosa level ≤1 mm; multivariate analysis, OR:0.599; $p = 0.024$).[72]

4.2.2 | Pooled various surgical therapies

Results from studies pooling different surgical treatment modalities, including resective and reconstructive approaches, are less consistent regarding the role of keratinized mucosa on surgical peri-implantitis treatment effectiveness.[64,66,73] From these, one retrospective analysis involving 40 patients with 68 implants treated with either resective or reconstructive approaches found that reduced keratinized mucosa width (<2 mm) exhibited significantly higher rates of Sup and lower marginal bone levels prior to the treatment.[73] However, following surgical treatment, no significant differences were observed between implants with sufficient (≥2 mm) or reduced (<2 mm) keratinized mucosa in terms of changes in BOP, PD reduction, or overall treatment success.[73] In contrast, another retrospective analysis investigating both resective and reconstructive therapeutic modalities identified that reduced keratinized mucosa width (<2 mm) significantly increased the risk for treatment failure (i.e., BOP and/or longitudinal bone loss ≥0.5 mm + PD ≥5 mm) when reconstructive treatment was performed (OR:13.9; $p = 0.02$),[64] but not when resective treatment was applied.[64] Supporting this, a recent sub-analysis of a multicenter RCT revealed that the absence of keratinized mucosa was significantly associated with residual BOP 12-months after non-submerged healing following surgical treatment for peri-implantitis, irrespective of whether adjunctive reconstructive measures were employed (OR:5.18).[66] Nonetheless, keratinized mucosa width showed no association with any of the other clinical parameters, including changes in PD, soft-tissue recession, marginal bone level alterations, and overall treatment success, regardless of the surgical approach used.[66]

Taken together, the clinical data suggest that the presence of keratinized mucosa appears to contribute to peri-implant health and favor the outcomes of surgical peri-implantitis treatment, particularly in cases involving reconstructive measures.

4.3 | Implant location

4.3.1 | Pooled various surgical therapies

Several studies have investigated whether implant location influences the outcomes of surgical peri-implantitis treatment, but evidence remains inconsistent. One retrospective analysis including both reconstructive and non-reconstructive treatment approaches, with a mean follow-up period of 3.5 years, reported a significantly

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lower risk of treatment failure (i.e., BOP and/or longitudinal bone loss ≥0.5 mm and/or PD ≥5 mm) at posterior implants compared with anterior ones (OR:0.22; $p=0.0017$), corresponding to a 78% reduction in failure risk.[64] When reconstructive therapy alone was considered, implants placed in the mandible were significantly more likely to show radiographic bone gain following surgery compared with those in the maxilla (OR:0.07; $p=0.021$),[64] whereas no association was observed in the group receiving resective treatment.[64]

Another retrospective study pooling various surgical treatment modalities, found that implants located in the maxilla were significantly associated with a higher risk of disease recurrence, that is, radiographic bone loss >1 mm, need for surgical retreatment, or implant loss/removal, in both univariate (OR:6.71; $p<0.001$) and multivariate logistic regression models (OR:9.69; $p=0.008$), after a mean follow-up of 3 years.[59] In addition, one retrospective analysis reported significantly higher implant loss rates after 12 months following reconstructive and non-reconstructive surgical treatment for the implants in molar regions compared with anterior implants (multilevel analysis: OR 3.21, $p=0.03$).[56]

4.3.2 | Reconstructive therapy

Opposing outcomes were reported in a 12-month RCT evaluating the adjunctive use of topical gaseous ozone therapy during reconstructive peri-implantitis surgery.[74] In this study, implants placed in the maxilla were positively associated with treatment success, that is, PD <5 mm, absence of BOP and/or Sup, no further bone loss, and radiographic defect fill ≥1 mm (OR:0.022; $p=0.031$).[74]

On the other hand, several other studies have failed to detect any significant influence of implant position (e.g., maxilla vs. mandible, anterior vs. posterior) on the outcomes of either surgical non-reconstructive and reconstructive peri-implantitis therapy.[43,58,62,75–76]

4.4 | Implant surface

Implant surface characteristics have been shown to influence the outcomes of non-reconstructive/resective surgical approaches for managing peri-implantitis.

4.4.1 | Non-reconstructive/Resective therapy

A 3-year follow-up study of non-reconstructive peri-implantitis treatment reported that implants with a non-modified (turned) surface exhibited greater PD reduction (2.90 ± 2.12 mm) and a lower prevalence of residual PD >5 mm (4 ± 12.9% of implants), compared with those with modified (rough) surfaces (2.67 ± 2.48 mm; 21 ± 23.3% of implants).[52] At 5 years, the same patient cohort showed that implants with modified surfaces carried a significantly higher risk of disease recurrence (i.e., bone loss ≥1.0 mm, surgical retreatment or implant removal/loss) compared with non-modified surface implants (multivariate analysis: OR:5.07, $p=0.01$).[76] As speculated by the authors, unfavorable outcomes of rough surfaces implants could be attributed to the difficulty of achieving complete surface decontamination during surgery, as well as their greater susceptibility to bacterial recolonization post-surgery, ultimately increasing the risk of disease recurrence over time.[76]

The aforementioned findings are in agreement with a 7-year retrospective analysis, which identified modified implant surface as a significant predictor of implant loss following non-reconstructive peri-implantitis therapy (HR: 4.5, $p=0.016$).[63] Likewise, another 2 to 11 year retrospective clinical investigation involving 50 implants treated with a resective surgical approach, implants with non-modified surfaces achieved superior treatment outcomes to those with modified surfaces, evidenced by significantly higher proportions of implants that were BOP-negative, either alone or in combination with PD ≤5 mm, and no bone loss ≥0.5 mm.[77]

4.4.2 | Reconstructive therapy

As demonstrated by the clinical study with 7-year of follow-up, implant surface roughness also influenced the outcomes of reconstructive peri-implantitis therapy. In particular, following a reconstructive treatment of intrabony defects using deproteinized bovine bone mineral with 10% collagen, treatment success, that is, PD ≤5 mm, no BOP or suppuration, and no further bone loss, was achieved in only 14.3% (2/14) of implants with rough (titanium-plasma-sprayed, TPS) surfaces, compared with 58.3% (7/12) of implants with moderately rough (sandblasted, acid-etched, SLA) surfaces.[78] Furthermore, considerably greater improvements in clinical parameters, including BOP, PD, and plaque scores, were observed at SLA implants than TPS implants.[78]

4.5 | Postoperative clinical parameters as predictor for disease recurrence

Several studies suggest that postoperative clinical parameters can serve as predictors of long-term stability after surgical non-reconstructive peri-implantitis treatment.

4.5.1 | Non-reconstructive/Resective therapy

In a 5-year longitudinal prospective study on non-reconstructive surgical therapy, implants displaying residual PDs of ≥6 mm at the 1-year follow-up after had significantly higher odds of disease recurrence, that is, radiographic bone loss ≥1.0 mm, the need for surgical retreatment, or implant loss/removal, compared with implants without such residual pockets (multilevel regression analysis: OR = 7.4; 95% CI: 2.8–19.3; $p<0.001$).[76] Disease recurrence/progression occurred in 80% of implants with residual PDs ≥6

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versus 29% of those without.[76] Similarly, radiographic marginal bone level at 1 year was positively associated with the risk of further disease deterioration (OR = 1.4; 95% CI: 1.1–1.7; $p = 0.01$).[76] Consistent outcomes were reported in another 5-year longitudinal investigation, which showed that the presence of residual PD ≥5 mm (at 3 or 4 sites) in patients treated with non-reconstructive surgical approach, significantly increased the risk of further disease progression ($p = 0.028$).[79] Notably, disease progression among implants with circumferential residual PDs increased from 23% during the first 5 years to 53% between the 5- and 10-year follow-ups.[80] Moreover, a retrospective analysis including 50 patients with 95 implants followed for 2.0 to 10.8 years after non-reconstructive surgical treatment, found that the probability of no further bone loss was 77% in the absence of BOP and 83% when PD was ≤5 mm at follow-up.[77] When both conditions were met (no BOP and PD ≤5 mm), the probability of no additional bone loss was 78%.[77]

In summary, residual peri-implant pockets, persistent BOP, and unfavorable bone levels after surgery strongly predict disease recurrence, underscoring the importance of achieving thorough disease resolution at early follow-up visits.

4.6 | Type of prosthesis

Prosthetic design has been identified as a potential factor influencing peri-implant health and surgical outcomes. Previous clinical studies have shown that implants supporting removable prostheses carry a higher risk of peri-implant disease compared with those restored using fixed reconstructions.[81–83]

4.6.1 | Pooled various surgical therapies

Two retrospective studies have further reported on the influence of prosthetic design on the treatment outcomes of surgical peri-implantitis therapy, following either non-reconstructive or reconstructive approaches.[56,59] In one study, after a mean follow-up of 30.80 ± 19.62 months, implants supporting overdentures had 3.6-fold higher odds of requiring surgical retreatment compared with those supporting single crowns (multilevel analysis, OR = 3.59, $p = 0.003$). In another retrospective analysis, disease recurrence (i.e., bone loss >1 mm, or the need for surgical retreatment, or implant loss/removal) was evaluated over a mean follow-up of 3.06 ± 1.52 years.[59] In this study, fixed complete prostheses were associated with significantly higher odds of recurrence (univariate analysis: OR = 9.45, $p = 0.024$) compared with single implant-supported crowns or fixed partial dentures; however, this association was not retained in the multivariate model.[59] The higher risk for disease recurrence associated with removable and full-arch restorations may be partly explained by the greater difficulty in achieving and maintaining effective plaque control compared with fixed prostheses. Furthermore, in the majority of cases, edentulism leading to full-arch restorations is a consequence of a history of periodontitis, which, as discussed in the previous section, may also favor disease recurrence following surgical peri-implantitis therapy.

4.7 | Macrophage polarization

Multiple studies have highlighted the pivotal role of macrophage polarization in the immunopathology of peri-implantitis.[84–86] During the onset and progression of peri-implant disease, monocytes are recruited in response to damage- or pathogen-associated signals and differentiate toward distinct functional macrophage subsets, most notably the classically activated M1 (pro-inflammatory) or the alternatively activated M2 (anti-inflammatory) phenotypes, depending on tissue-specific microenvironmental stimuli.[86,87] M1 macrophages are a major source of pro-inflammatory cytokines, including interleukin (IL)-1β, IL-6, IL-12 and tumor necrosis factor (TNF)-α, and are key contributors of persistent inflammation and peri-implant tissue breakdown.[88,89] In contrast, M2 macrophages promote inflammation resolution and tissue repair through the expression "anti-inflammatory" mediators such as arginase (ARG)-1, IL-4, IL-10, IL-35, and transforming growth factor (TGF)-β1, while supporting the clearance of apoptotic cells and maintaining mucosal homeostasis.[90–93]

Consistent with this paradigm, peri-implantitis lesions—primarily characterized by immunohistochemical and immunofluorescent analyses—exhibit a marked shift toward M1 polarization, while presenting an increased M1/M2 detection ratio in peripheral blood, as assessed by flow cytometry using phenotype-specific surface markers, which positively correlates with plaque index, bleeding on probing, and probing depth.[94] Clinical data further demonstrate a progressive increase in M1 expression with peri-implantitis progression, where sites showing marginal bone loss exceeding 50% of the implant length present a significantly higher percentage of $CD80^+$ M1 macrophage immunodetection than moderate or early peri-implantitis lesions.[85] Experimental models of peri-implantitis corroborate this M1-predominant response,[95] characterized by upregulated expression of the M1-associated marker inducible nitric oxide synthase (iNOS) and the cytokines IL-1β, IL-6, and TNF-α, together with the reduced expression of the M2-associated marker CD206 and the inflammation–resolving mediators Arg-1 and IL-10 within peri-implant lesions.[96,97] In this context, the therapeutic inhibition of M1 polarization, attenuating peri-implant inflammation and bone loss, provides additional relevant evidence that supports the central pathogenic role of M1 macrophages.[94,95] Suppressing M1 activity leads to reduced peri-implant tissue levels of IL-1β, IL-6, and TNF-α expression, while upregulating osteogenic markers such as alkaline phosphatase, Runx2, and osteocalcin. This phenotypic shift not only dampens chronic inflammation but also limits bone resorption across multiple peri-implantitis models.[95,98–99]

A retrospective clinical study further investigated macrophage polarization in granulation tissue samples collected from 14 patients undergoing surgical combined therapy for peri-implantitis.

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Immunohistochemical analysis revealed that macrophages occupied 15.3% ± 7.0% of the granulation tissue area, with M1 and M2 subsets representing 7.1 ± 4.1% and 5.5 ± 3.7%, respectively, with no significant difference between them ($p = 0.16$). Six months after combined surgical therapy (i.e., implantoplasty combined with reconstructive treatment) patients exhibited significant improvements in clinical parameters, including BOP, PD, and Sup.[100] Linear regression analysis further revealed that higher baseline M1 expression was associated with greater PD reduction and increased soft-tissue recession after 6 months.[100] Complementarily, a study analyzing differential cytokine expression profiles from 48 peri-implant tissue samples by means of real time quantitative PCR, reported significantly higher M1-associated *IL12* and *TNFA* mRNA expression levels in severe cases of peri-implantitis, than early peri-implantitis and mucositis ($p < 0.05$).[101] Conversely, M2-associated *IL4 mRNA* exhibited the highest expression levels in healthy peri-implant tissues, with progressively lower levels in mucositis, early peri-implantitis and severe peri-implantitis samples ($p < 0.05$).[101] Collectively, these findings suggest that the M1/M2 balance plays a critical role not only in peri-implant disease severity but also in the healing response following surgical peri-implantitis treatment.[100]

In a more recent prospective investigation, peri-implantitis lesions and peri-implant crevicular fluid from 30 patients treated with reconstructive therapy employing volume-stabilized xenogenic bone filler and a collagen membrane were analyzed after 6 and 12 months (unpublished data; Figure 6). The mRNA expression profiles showed elevated levels of M1-associated markers — including IL-1β, IL-6, IL-12p40, iNOS, and TNF-α— as well as the M2-associated marker CD206, together with reduced IL-4 expression, compared with healthy controls ($p < 0.05$). Higher M1/M2 immunodetection ratios were associated with greater vertical defect depth (VDD), PD, and BOP, whereas increased expression of M2-associated ARG-1 and IL-10 protein levels in crevicular fluid correlated with more favorable clinical outcomes. Interestingly, a sub-analysis showed that non-responders—that is, patients exhibiting no reductions in PD or BOP after 12 months—had higher iNOS levels at 12 months than responders, despite reductions from baseline ($p < 0.05$) (unpublished data). In contrast, ARG-1 increased significantly in responders but not in non-responders, resulting in a significant between-group difference ($p < 0.05$) (unpublished data). Together, these findings indicate that macrophage polarization status may influence not only peri-implantitis disease progression, but also the likelihood of achieving a favorable clinical response to surgical treatment.

5 | PATIENT-RELATED FACTORS

5.1 | History of periodontitis

Robust evidence from longitudinal and cross-sectional studies indicates that a history of periodontitis represents a major risk factor for peri-implantitis.[102] A meta-analysis reported that patients with periodontitis history had significantly lower implant survival rates and considerably increased risk of peri-implantitis ≥5 years after placement compared with periodontally healthy patients restored with implant-supported fixed partial dentures.[103] Similarly, a recent 20-year follow-up prospective clinical investigation including 84 patients with 172 implants found that deep pockets (PD ≥6 mm) and marginal bone loss ≥3 mm occurred significantly more frequently among patients with history of periodontitis.[104] Notably, non-compliant periodontally compromised patients experienced more biological complications than compliant patients undergoing individually tailored supportive care.[104]

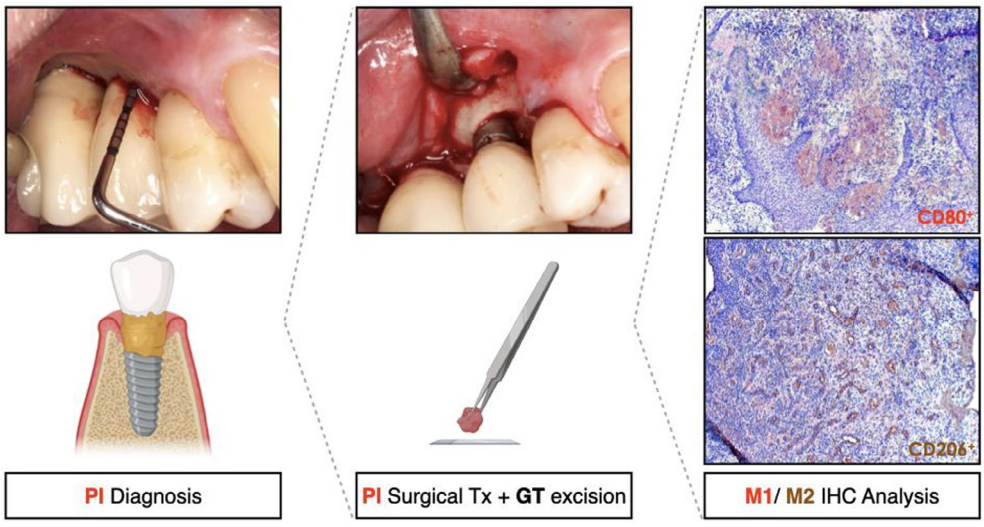


FIGURE 6 Macrophages polarization analysis workflow. Left panels: Schematic illustration and intraoral picture depicting an implant affected by peri-implantitis, highlighting bleeding on probing as an inflammatory sign for diagnosis. Central panels: Intraoperative view of peri-implantitis surgical treatment (TX) and access for granulation tissue (GT) excision for subsequent immunohistochemical analyses. Right panels: Immunohistochemical staining for CD80 [AEC chromogen (red)] and CD206 [DAB chromogen (brown)] for the detection of M1 and M2 macrophages within the peri-implantitis lesion inflammatory infiltrate, respectively.

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5.1.1 | Non-reconstructive/Resective therapy

Contradictory clinical data exist regarding the role of periodontitis history on the outcomes of surgical peri-implantitis treatment. In particular, one prospective longitudinal study with a 5-year follow-up after non-reconstructive peri-implantitis treatment showed no significant association between treatment failure (i.e., bone loss >1.0 mm, surgical retreatment, or implant removal/loss) and a history of periodontitis, despite most patients being periodontally compromised (at 1 year: 84.5%, at 5 years: 84.9%).[76] Consistently, a sub-analysis of two randomized clinical trials (RCTs) applying a resective surgical approach found that history of periodontitis (reported in 44% [32/72] of patients) was not associated with treatment failure after 1 year (i.e., implant removal/loss, progressive bone loss ≥1 mm + BOP at 3 to 4 sites/Sup, PD ≥5 mm).[58] It should be noted, however, that history of periodontitis was primarily self-reported by patients rather than clinically diagnosed.[58] In contrast are the finding of one retrospective analysis, which after a mean follow-up period of 3.5 years reported a significantly higher risk of unsuccessful treatment outcome (i.e., BOP, and/or bone loss ≥0.5 mm, and /or PD >5 mm) in patients with a history of periodontitis treated with a resective treatment modality (univariate analysis: OR:3.07, $p = 0.04$), but not in those treated with reconstructive procedures (univariate analysis: OR:0.75, $p = 0.71$).[64]

5.1.2 | Reconstructive therapy

In the context of reconstructive peri-implantitis therapy, a 3-year RCT employing bovine bone mineral with either a collagen or concentrated growth factor membranes to treat peri-implantitis-related intrabony defects found that periodontitis history significantly reduced the odds of treatment success (i.e., no additional bone loss and PD ≤5 mm, multivariate analysis: odds ratio (OR): 3.5, $p = 0.033$).[43] However, in the same patient cohort at 7-year follow-up, no association between history of periodontitis and a successful treatment outcome was confirmed (i.e., BOP ≤1 site, PD ≤5 mm, no additional bone loss, no Sup, peri-implant soft-tissue deficiency ≤1 mm, no surgical retreatment).[44] Likewise, a sub-analysis of two RCTs comparing submerged vs. non-submerged healing after reconstructive peri-implantitis therapy reported no significant influence of periodontitis history on changes in marginal bone levels, PD and BOP, despite most patients having a history of periodontitis (87%, 40/46).[48]

5.1.3 | Pooled various surgical therapies

Conversely, two retrospective analyses combining various surgical approaches, including resective, reconstructive, and combined therapies for peri-implantitis reported a stronger negative effect of periodontitis history. Specifically, patients with Grade C periodontitis had fourfold higher risk of disease recurrence (i.e., PD ≥6 mm + profuse BOP) than Grade A patients (OR:4.6, $p = 0.01$) after 12 months.[75] In line with this, another retrospective study including 34 patients (70% with history of periodontitis) with 103 implants also found a significantly increased risk of treatment failure (i.e., radiographic bone loss ≥1 mm, need for surgical retreatment, or implant loss/removal) after a mean follow-up of 3.06 ± 1.52 years (OR:4.62, 95% CI: 1.14–18.75, $p = 0.032$).[59]

Taken together, while a history of periodontitis is a well-established risk factor for peri-implantitis development, its influence on surgical treatment outcomes remains less conclusive. Indeed, it should be acknowledged that the heterogeneity between study designs, inconsistent definitions of periodontitis in the majority of studies, and differing success criteria may likely explain the variability across the reported outcomes.

5.2 | Smoking

The evidence regarding smoking as a risk factor or indicator for peri-implantitis remains inconclusive.[102] Whereas some studies have failed to demonstrate a significant association between smoking status and peri-implantitis,[105–107] others reported a clear correlation between smoking and higher risk for peri-implantitis.[61,108–111] A recent meta-analysis based on five prospective cohort studies revealed a strong association between smoking and the risk of peri-implantitis at both the implant level (risk ratio [RR]: 2.04, $p < 0.0001$) and the patient level (RR: 2.08, $p = 0.01$).[112] Supporting these findings, a cross-sectional study analyzing macrophage polarization in granulation tissue biopsies showed that samples harvested from smokers (≥10 cigarettes/day) requiring surgical peri-implantitis treatment had a significantly higher proportion of M1 pro-inflammatory macrophages (70.2% vs. 25.1% in non smokers), along with more severe peri-implant tissue destruction.[113] These results suggest that smoking may promote a pro-inflammatory immune response, potentially exacerbating peri-implant tissue breakdown and compromising treatment outcomes.

5.2.1 | Non-reconstructive/Resective therapy

Clinical data on smoking's influence on surgical treatment success in peri-implantitis are conflicting. A retrospective analysis involving 45 patients (17.8% smokers) treated with non-reconstructive surgery over a mean follow-up period of 6.4 years found smoking to significantly increased the risk of further disease progression (multivariate analysis, OR:6.36, $p = 0.023$).[62] Similarly, another study identified smoking as a prognostic factor for treatment failure (i.e., implant loss or progressive bone loss ≥1 mm + Sup/BOP at 3–4 sites, PD ≥5 mm) after 1 year of non-reconstructive surgery (multivariate analysis, OR:3.82, $p = 0.044$).[58] However, other studies with follow-up periods ranging from 1 to 5 years reported no significant differences in clinical outcomes between smokers and nonsmokers following non-reconstructive peri-implantitis therapy.[60,76] Likewise, studies

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assessing the efficacy of reconstructive peri-implantitis therapy with 5 to 7 years follow-up of failed to identify smoking as a significant predictor of clinical treatment outcomes.[38,44,48]

5.2.2 | Pooled various surgical therapies

Similar contradictions also arise in studies pooling different surgical approaches. Specifically, a retrospective study including 245 patients found that nonsmokers were significantly more likely to achieve treatment success (i.e., BOP/Sup, PD <5 mm, and stable marginal bone levels), than smokers ($p=0.003$), and that smoking was significantly associated with treatment failure (i.e., no treatment success, $p=0.0023$).[114] In a secondary analysis of a RCT comparing reconstructive and non-reconstructive approaches, smokers presented significantly higher residual PD values after 12 months (mean difference 0.6 mm), with smoking emerging as a significant predictor of poor clinical outcome (multivariate analysis, OR:0.60, $p<0.05$), regardless the treatment modality.[66] Supporting this, a retrospective analysis showed that smokers had a fourfold increased risk of disease recurrence (i.e., PD ≥6 mm + profuse BOP) 12 months post-surgery, independent of the applied surgical technique (OR:4.49, $p=0.003$).[75] Nevertheless, other studies with mean follow-up periods of ~3 to 3.5 years reported no association between smoking status and clinical outcomes after either reconstructive or non-reconstructive peri-implantitis therapy.[23,59,64] In light of these conflicting results, no definitive conclusion can be drawn regarding the impact of smoking on the success of surgical peri-implantitis treatment.

5.3 | Gender

The potential influence of patients' gender on the outcomes of surgical peri-implantitis treatment has been investigated in some studies, with inconsistent findings. In particular, a 12-month RCT reported that female patients had a significantly higher probability of disease resolution (i.e., ≤1 site with BOP, absence of Sup, and PD ≤5 mm) following reconstructive peri-implantitis therapy (multivariate analysis: OR:5.56, $p=0.02$).[37] On the contrary, two retrospective studies pooling various surgical treatment approaches reported opposite results.[11,75] According to one analysis, female patients, comprising the majority of the study population (103/132), were approximately one-third more likely to experience disease recurrence (i.e., PD ≥6 mm and profuse BOP) compared with male patients (univariate analysis, OR:0.34, $p=0.01$).[75] However, gender did not remain as a significant factor in the multivariate model.[75] Similarly, another retrospective study found that the female gender significantly increased the risk of treatment failure (i.e., radiographic bone loss >1 mm, need for surgical retreatment, or implant loss/removal) (univariate analysis, OR:12.60, 95% CI: 1.16–26.14, $p=0.032$), but this association also did not remain significant after adjustment in the multivariate model.[59] Otherwise, several other studies evaluating both reconstructive and non-reconstructive treatment modalities failed to identify any significant association between patient gender and therapeutic outcomes of surgical treatment of peri-implantitis.[38,43,48,64,80]

5.4 | Postoperative adherence to supportive therapy

One of the most critical determinants of long-term success in peri-implantitis management is the patient's adherence to supportive postoperative care (maintenance therapy). Among the included studies, relatively few explicitly reported details about maintenance regimens. Thirty studies clearly stated that patients were enrolled in a structured supportive care program after surgery, and from these only 3 studies did not specify the frequency of maintenance visits. Nevertheless, abundant evidence from the literature underscores that inadequate plaque control and infrequent maintenance are major risk factors for peri-implantitis and its recurrence.[102]

A 5-year longitudinal study demonstrated that the incidence of peri-implantitis was markedly lower in patients who adhered to supportive maintenance therapy compared with those who did not (18% vs. 44%).[115] Similarly, over a 20-year observation period, non-compliant periodontally compromised patients more frequently exhibited PDs exceeding 6 mm and reduced implant survival rates compared with compliant individuals.[104] Moreover, preoperative plaque levels also appear to significantly influence the outcome of surgical therapy for peri-implantitis.[37] Specifically, lower baseline modified Plaque Index (mPI) values were statistically associated with treatment success (i.e., absence of profuse BOP, Sup, PD ≤5 mm, and no radiographic bone loss ≥1 mm), with each one-unit increase in mPI being associated with reducing the odds of treatment success by 87% (odds ratio [OR]: 0.13, $p=0.006$).[37]

Several studies have confirmed that poor postoperative plaque control negatively affects surgical therapy for peri-implantitis outcomes. Based on an univariate regression analysis, the presence of plaque 12 months after non-reconstructive peri-implantitis surgery significantly increase the odds of treatment failure (i.e., implant loss, or progressive bone loss ≥1 mm + BOP at 3 to 4 sites/Sup + PD ≥5 mm) (OR:2.92, $p=0.042$).[58] Furthermore, over a mean 6.5 year follow-up after non-reconstructive surgical therapy, compliant patients —attending all scheduled recall visits— had a significantly lower treatment failure rate, defined as implant removal or further bone loss ≥1.0 mm, compared with non-compliant patients (i.e., attending fewer than 50% of recall visits; multivariate analysis: OR:0.15; $p=0.019$).[62] Interestingly, treatment failure in non-compliant patients were comparable to those of erratic compliers, attending more than 15% of recall visits ($p=0.58$).[62] Likewise, a prospective case series assessing resective surgical approach showed that presence of plaque at implant sites after 6-months was

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significantly associated with treatment failure (PD ≥6 mm + BOP/Sup).[61]

In the context of reconstructive peri-implantitis therapy, patient compliance also plays a critical role. A 12-month case series reported that unresolved peri-implantitis sites exhibited mPI values 18.3% higher than those observed at resolved sites, which were defined as having no BOP or Sup upon gentle probing.[116] A recent 3-year RCT pointed to a tendency toward more favorable clinical outcomes of reconstructive peri-implantitis treatment in fully compliant patients (supportive therapy 2x year) versus partially compliant ones (PD ≤5 mm, no BOP/Sup, buccal recession ≤1 mm: 32% vs. 13%).[40]

A retrospective analysis pooling both reconstructive or resective peri-implantitis therapies confirmed that lack of postoperative compliance and inadequate oral hygiene are strongly associated with reduced treatment success (i.e., no BOP or Sup and PD ≤4 mm).[65] Specifically, patients with unacceptable compliance, defined as failure to follow the recommended treatment plan and hygiene instructions, showed significantly lower treatment success rates, compared with compliant patients (OR:5.0, $p < 0.05$).[65] In addition, patients exhibiting inadequate oral hygiene at follow-up, as indicated by a plaque index ≥20%, showed significantly reduced treatment success (OR:2.9, $p < 0.05$).[65] More recently, a sub-analysis of a RCT identified plaque presence at 6 weeks postoperatively as a significant predictor of residual PDs >5 mm (coefficient: 1.00, 95% confidence interval [CI]: 0.30–1.70) and persistent BOP (OR:5.18, 95% CI: 1.14–10.56) at 12 months following either reconstructive or non-reconstructive surgical therapy.[66]

In summary, the aforementioned clinical data collectively underscore the importance of both self-performed and professionally administered biofilm control measures in preventing disease recurrency at the implant site following surgical treatment of peri-implantitis.

6 | CONCLUDING REMARKS

Surgical peri-implantitis treatment outcomes are influenced by a multitude of factors spanning surgical technique, local site characteristics, and patient-level behaviors. Key conclusions from the current evidence include:

Surgical factors

- The adjunctive use of reconstructive (regenerative) measures generally improves radiographic defect fill and may help maintain soft-tissue levels, but no clear advantage has been shown in resolving clinical signs of inflammation compared with non-reconstructive open-flap debridement.
- Reconstructive protocols that incorporate slowly resorbing bone substitutes (e.g., xenografts) tend to yield more favorable outcomes than those using autogenous bone, particularly in terms of bone fill and possibly pocket reduction.
- There is insufficient evidence to support the routine use of biologics (growth factors, EMD, PRF, etc.) as adjuncts to surgical therapy, as they have not demonstrated significant additional benefits in controlled clinical studies.
- Contained intrabony defects (3- and 4-wall) show considerably more favorable outcomes with reconstructive therapy compared with non-contained 1- or 2-wall defects.
- No clear clinical benefits could be observed for the adjunctive use of a membrane in reconstructive therapy. Their application should be weighed against defect morphology (more justified in 1–3 wall defects and possibly unnecessary in 4-wall defects).
- The effectiveness of peri-implantitis surgery is tied to the surgeon's experience and technique, with experienced surgeons achieving higher success rates.
- No definitive conclusions can be drawn regarding the flap design or postoperative healing mode (submerged versus non-submerged). While submerged healing seems to improve clinical outcomes following reconstructive therapy in some studies, practical constraints and complication risks must be considered on a case-by-case basis.
- Similarly, the adjunctive use of systemic antibiotics remains controversial. While some short-term benefits have been observed (especially in reducing suppuration and early reinfection), their routine use is not currently recommended given inconsistent results and concerns over antibiotic resistance.

Local implant-site factors:

- Greater initial disease severity is consistently associated with worse treatment outcomes. Extensive baseline bone loss, deep initial probing depths, and presence of suppuration are strong negative predictors of success.
- Residual pockets (PD >5 mm) or persistent bleeding on probing following surgery have been identified as major determinants compromising surgical success.
- Rough-surfaced implants have higher odds of recurrence and complications post-surgery compared with machined implants, likely due to the difficulty in decontaminating rough surfaces and their tendency to accumulate plaque.
- The presence of keratinized mucosa appears to improve the outcomes of reconstructive peri-implantitis therapy.
- While the effect of implant location (jaw and region) on treatment outcomes is poorly investigated, there is some limited evidence that maxillary or anterior implants might have a higher risk of re-treatment in certain populations.
- Emerging evidence suggests that the local immune response, particularly macrophage polarization (M1 vs. M2 phenotype), might influence healing. Modulation of this response could become a therapeutic target in the future.

Patient-related characteristics:

- Current clinical evidence highlights the critical role of patients' adherence to supportive postoperative therapy in maintaining surgically obtained outcomes.
- Findings regarding the impact of a history of periodontitis, smoking status, and gender remain inconclusive.

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- The scarcity of clinical data did not allow us to evaluate the influence of patients' systemic health status on the effectiveness of surgical peri-implantitis treatment.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there are no conflicts of interest regarding the publication of this paper.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Ausra Ramanauskaite https://orcid.org/0000-0002-1649-1882
Wenije Zhou https://orcid.org/0000-0001-8158-253X
Emilio A. Cafferata https://orcid.org/0000-0001-9243-1382
Frank Schwarz https://orcid.org/0000-0002-5873-9903

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